TRAINS DETAINED & TCA

Detention on Running Lines & Track Circuit Actuators

Revision Guide — EMR Training | *GERT8000-S4 Issue 5* | *GERT8000-TW5 Issue 14* | *RS521 Issue 9*

⚠ All rule book content sourced exclusively from GERT8000-S4, GERT8000-TW5 and RS521 ⚠

(Section 2.4 is the one exception — labelled there as classroom material, not Rule Book text)

1. Trains or Shunting Movements Detained on Running Lines (GERT8000-S4)

People responsible: driver, signaller, shunter.

1.1 Contacting the signaller — standard arrangements

When to contact the signaller

If your train is detained on a running line at a signal at danger, or without a movement authority (MA), you must contact the signaller as soon as possible.

- You may wait up to 2 minutes before contacting the signaller IF you can see an obvious reason for the signal being at danger / no MA, e.g. the section ahead is occupied, or a conflicting movement is being made.
- If the signaller has told you to wait for the signal to clear (or for an MA), you must contact them again every 5 minutes, unless told otherwise.

Key numbers: *Up to 2 minutes before first contact (if obvious reason) — then re-contact every 5 minutes if told to wait.*

How to contact the signaller

Escalation order if the train radio cannot be used:

Step	Method
1	Train radio (always the first method)
2	Signal post telephone (SPT) — unless limited clearance prevents this
3	Mobile phone (if SPT not provided / failed, or not available)
4	Telephone at another signal, a lineside telephone, or go to the signal box

When speaking to the signaller

- Make sure you are speaking to the correct signaller.
- Make sure the signaller clearly understands which signal/block marker your train is at, and on which line.
- If detained without an MA and not at a signal/block marker, reach a clear understanding of your train's location and line.
- Tell the signaller your train reporting number.

When speaking to the driver (signaller's duty)

- Tell the driver the reason for the delay.
- Instruct the driver to ‘wait for the signal’ or ‘wait for an MA’.

Driver being conducted

If you don't have the required route knowledge and are accompanied by a conductor driver, the conductor driver must contact the signaller and pass on any instructions to you.

1.2 Contacting the signaller — non-standard arrangements

Telephone sign with a number displayed

If a number is shown on the telephone sign at the signal, or a waiting time is shown in the Sectional Appendix for that area, you must contact the signaller within that number of minutes (instead of 'as soon as possible').

White diamond sign with a telephone number

If a white diamond sign shows a telephone number and you cannot contact the signaller from the cab by any means, you must only leave the cab to use another telephone:

- in an emergency, or
- if another driver, or a competent person, has confirmed the signaller has blocked the adjacent line and it is safe to get down.

Signaller's duty if the driver can't be contacted and the signal/MA can't be cleared/issued: instruct a train passing on another line to stop opposite the detained train's cab and relay the message. If no train is available, stop trains on the adjacent line and send a competent person to tell the driver it's safe to get down. Normal working on the adjacent line must not resume until the signaller is sure the detained train has moved on.

During poor visibility

On lines other than TCB or ERTMS, if the signal has no white diamond sign and visibility is less than 180 metres (approx. 200 yards), you must contact the signaller immediately.

If the signal does have a white diamond sign, and you have to use another telephone or go to the signal box, you must do so within 10 minutes.

Trains conveying sensitive traffic

If your train is a block train of dangerous goods, or a mail/postal train, you must contact the signaller immediately (use an SPT only if radio/mobile failed).

Signaller's duty: if the reason for the signal being at danger (or no MA) cannot be identified, and the train is dangerous goods or mail/postal, this must be treated as suspicious and the police called immediately.

1.3 Limited clearance at signal post telephones

Limited clearance warning sign (no diamond/roundel)

Where there's a limited clearance warning sign but no white/yellow diamond with an 'X', you may use the telephone — you're in a position of safety and your train provides protection.

Yellow/white diamond with letter 'X', or yellow roundel on the telephone cabinet

You must NOT normally leave the cab to use the SPT. You may only do so:

- in an emergency, or

- if another driver, or a competent person, has confirmed the signaller has blocked the adjacent line and it's safe to get down.

Signaller's duty mirrors the white diamond procedure above: relay via another train if possible, otherwise stop adjacent-line trains and send a competent person; don't resume normal working until sure the train has moved on.

1.4 Shunting movement detained on a running line

- Driver: if a shunting movement has been detained an unusually long time, remind the signaller in the quickest way possible — this may mean going to the signal box yourself, or sending the shunter.
- Shunter: must go to the signal box to remind the signaller if instructed to do so by the driver, when the movement has been detained an unusually long time.

2. Track Circuit Actuators — TCA (GERT8000-TW5, Section 23)

People responsible: driver, signaller, train preparer.

Note: *These instructions do NOT apply to an on-track machine (OTM) being hauled dead.*

2.1 Starting a journey from a maintenance depot

You must NOT allow a train to start a journey if the TCA:

- is isolated on any vehicle, or
- the isolating switch is unsealed, or
- the warning light indicates a system fault.

2.2 Starting a journey from somewhere other than a maintenance depot

A train MAY start a journey with one or more defective/isolated TCAs, as long as the minimum working requirement below is met:

Train length	Minimum requirement
1–2 vehicles	At least one working TCA on the train
3+ vehicles	At least one working TCA on either of the first two vehicles, AND at least one working TCA on either of the last two vehicles

You must first tell the train operator's control.

A train that does NOT meet this requirement may still enter service if there is at least one working TCA on the train AND the train operator's control has given authority.

OTM exception: *An OTM may start a journey with a defective TCA, but only to travel to a maintenance depot for repair, or to/from an engineering possession directly. The signaller must be told the OTM cannot be relied upon to operate track circuits.*

2.3 During a journey

a) When the train can continue normally

If one or more TCAs become defective during the journey, the train may continue normally as long as the same minimum working requirement (table above) is still met.

- Tell the train operator's control at the first convenient opportunity.
- Carry out any instructions given.

(These instructions don't apply if it's necessary to isolate the TCA on an OTM before it can work in a protection zone.)

b) When the train can continue normally with authority

If the train does NOT meet the minimum working requirement:

- Tell the signaller immediately which vehicle is defective.
- Do not move the train until instructed.
- Carry out the instructions given.

The train may then continue its journey as long as there is at least one working TCA on the train AND the driver is told the train operator's control has given permission.

c) When the train cannot continue normally

Applies when the train doesn't meet the minimum requirement and cannot be given authority under (b).

- Driver: do not move the train until instructed; carry out the instructions given.
- Signaller: must make sure the signal protecting the train stays at danger (or, on ERTMS, keep the route closed) and signal the train as shown in TS1 regulation 12.
- Except at an automatic half-barrier crossing (AHBC) fitted with treadles, the driver must be instructed to approach any automatic level crossing, or barrow/foot crossing with white light indications, at caution and not pass over until sure it's safe.
- If told to approach a level crossing at caution, the driver must sound the warning horn continuously from where caution travel starts until the front of the train is on the crossing.
- Once given authority to proceed, the driver may resume normal speed.

2.4 Supplementary: How a TCA actually works

Source note: *This subsection comes from the classroom slides (PPTX), not from TW5 itself. TW5 states the operational rule only - it does not explain the underlying electrical theory. Included here for background understanding.*

A track circuit is normally completed by a train's own wheels and axles, which let a small electrical current flow through the rails back to a relay. While that current flows, the relay knows the track is occupied; when it stops, the relay knows the track is clear.

A TCA (Track Circuit Actuator) is fitted between a pair of wheelsets on a vehicle. It raises the voltage at the wheel-to-rail contact point on either side of it.

- This lets a low-voltage track circuit operate as reliably as a higher-voltage type.
- Over time, rust can form a thin film on the wheels and rails that stops the track circuit's small current flowing properly. The TCA's higher voltage is enough to break down that rust film, allowing current to flow and the relay to de-energise correctly.

In short: the TCA doesn't communicate anything on its own - it physically helps the train's own wheels reliably complete the track circuit. That's why the rule in TW5 §23 cares about how many TCAs are working, and where on the train.

3. RS521

RS521 (Colour light & general signalling principles) was checked but contains no provisions on ‘trains detained’ or TCA — these topics are covered only in GERT8000-S4 and GERT8000-TW5 §23 respectively. Nothing from RS521 has been used in this guide for that reason.

Quick Recap

- S4: contact signaller ASAP if detained at red/no MA — 2 min grace if obvious reason, then re-contact every 5 min if told to wait.
- S4: contact order — radio → SPT (unless limited clearance) → mobile → another telephone/signal box.
- S4: poor visibility (non-TCB/ERTMS, no white diamond) <180 m — contact immediately.
- S4: dangerous goods / mail trains — contact immediately; unexplained danger signal = treat as suspicious, call police.
- TCA: depot start banned if isolated on any vehicle / switch unsealed / warning light fault.
- TCA: minimum working rule — 1–2 vehicles need 1 working TCA; 3+ vehicles need 1 in the front two AND 1 in the back two.
- TCA: doesn't meet minimum + no authority → signal kept at danger, approach crossings at caution sounding horn continuously (except AHBC with treadles).

Q&A Self-Test

Cover the answers and test yourself. All answers are drawn directly from GERT8000-S4 and GERT8000-TW5 §23.

Q1. How quickly must a driver contact the signaller if detained at a signal at danger or without a movement authority (MA)?

A: As soon as possible — though the driver may wait up to 2 minutes first if there is an obvious reason (e.g. the section ahead is occupied, or a conflicting movement is being made).

Q2. If told to wait for the signal to clear or for an MA, how often must the driver re-contact the signaller?

A: Every 5 minutes, unless given other instructions.

Q3. What is the first method a driver must use to contact the signaller?

A: The train radio.

Q4. If the train radio cannot be used, what is the next method?

A: The signal post telephone (SPT) — unless limited clearance at the telephone prevents this.

Q5. If there is no SPT, or it has failed, what should the driver try next?

A: A mobile phone, if available.

Q6. If the driver still cannot contact the signaller after radio, SPT and mobile, what options are left?

A: Use the telephone at another signal, use a lineside telephone, or go to the signal box.

Q7. What three things must a driver confirm when speaking to the signaller?

A: That they are speaking to the correct signaller; that the signaller clearly understands which signal/block marker and line the train is at (or, if not at one, a clear understanding of the train's location); and they must give the signaller their train reporting number.

Q8. On a non-TCB/ERTMS line with no white diamond sign at the signal, at what visibility must the driver contact the signaller immediately?

A: Less than 180 metres (approximately 200 yards).

Q9. What must a driver of a block train of dangerous goods, or a mail/postal train, do if detained?

A: Contact the signaller immediately.

Q10. What must the signaller do if the reason for the danger signal/no MA cannot be identified and the train is dangerous goods or mail/postal?

A: Treat it as suspicious and call the police immediately.

Q11. When may a driver leave the cab to use an SPT marked with a yellow/white diamond with an 'X', or a yellow roundel on the telephone cabinet?

A: Only in an emergency, or if another driver or a competent person confirms the signaller has blocked the adjacent line and it is safe to get down.

Q12. What must a shunter do if instructed by the driver, when a shunting movement has been detained an unusually long time?

A: Go to the signal box to remind the signaller.

Q13. Under what three conditions must a train NOT be allowed to start a journey from a maintenance depot, due to the TCA?

A: If the TCA is isolated on any vehicle, the isolating switch is unsealed, or the warning light indicates a system fault.

Q14. What is the minimum working TCA requirement for a train of 1–2 vehicles starting a journey from somewhere other than a depot?

A: At least one working TCA on the train.

Q15. What is the minimum working TCA requirement for a train of 3 or more vehicles?

A: At least one working TCA on either of the first two vehicles, AND at least one working TCA on either of the last two vehicles.

Q16. Under what circumstances may an OTM start a journey with a defective TCA?

A: Only to travel to a maintenance depot for repair, or directly to/from an engineering possession — and the signaller must be told the OTM cannot be relied upon to operate track circuits.

Q17. If a TCA becomes defective during a journey but the minimum working requirement is still met, what must the driver do?

A: Tell the train operator's control at the first convenient opportunity, and carry out any instructions given.

Q18. If the minimum working requirement is NOT met after a TCA becomes defective during a journey, what must the driver do immediately?

A: Tell the signaller immediately which vehicle is defective, and not move the train until instructed to do so.

Q19. When the train doesn't meet the minimum requirement and cannot be given authority to continue, how must the driver approach an automatic level crossing (except an AHBC fitted with treadles)?

A: At caution, not passing over until sure it is safe, sounding the warning horn continuously from where caution travel starts until the front of the train is on the crossing.